Principal Components of US Sector Prices and Macroeconomic Data*

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1 Introduction

The relation between macroeconomic conditions and stock prices matters to investors, financial institutions and policymakers alike. Chen et al. (1986) show that macroeconomic variables, among them industrial production, changes in the risk premium, twists in the yield curve and inflation, are priced in US stock returns. Examining this relation directly is not trivial, since macroeconomic indicators are numerous and strongly correlated with one another, as are the sectors that make up the stock market. The high dimensionality of both sets and their multicollinearity obscure the contribution of any single variable. Principal component analysis (PCA) addresses the problem through an orthogonal transformation that converts correlated variables into uncorrelated principal components (PCs), ordered by the share of variance each explains.

This study asks how many components summarize 10 US macroeconomic indicators and 11 sector prices, and what those components represent. We apply PCA to 10 monthly US macroeconomic indicators and to the daily prices of the 11 Select Sector SPDR exchange-traded funds (ETFs). The two PCs in both datasets explain 78.26% of the macroeconomic variance and 93.62% of the sector price variance. The first macroeconomic component captures the joint movement of inflation and consumption measures, and the second sets wage growth and core inflation against durable goods orders and industrial production. The first sector component is a market-wide factor that energy loads least on, and the second is an energy factor.

We employ principal component regression (PCR) (Massy, 1965) to analyze how much of the S&P 500 return the sector components explain. The regression on the first two sector components explains 97.14% of the variance of the S&P 500 return in the training sample and attains a lower error in the testing sample than the regression on the first component alone.

Section 2 describes the data and their correlations. Section 3 presents PCA, the selection rules and PCR. Section 4 selects and interprets the components. Section 5 reports the results of PCR. Section 6 concludes.[1]

2 Data

Macroeconomic data Table 1 lists the 10 monthly US indicators, observed from January 2010 to September 2023, a total of 165 months. Eight indicators enter as year-on-year growth rates, $X_t/X_{t-12} - 1$, while the unemployment rate and the trade balance enter as levels.

Sector and index data The 11 Select Sector SPDR ETFs partition the S&P 500 into materials (XLB), communication services (XLC), energy (XLE), financials (XLF), industrials (XLI), technology (XLK), consumer staples (XLP), utilities (XLU), health care (XLV), consumer discretionary (XLY) and real estate (XLRE). We use daily adjusted closing prices for the 11 ETFs and the S&P 500 index. The training sample runs from 19 June 2018 to 30 December 2022, and the testing sample from 3 January 2023 to 2 November 2023. The PCA of sector prices uses the training prices, while PCR uses daily log returns $r_t = \ln(P_t/P_{t-1})$.

*This is a class project for the Chinese University of Hong Kong's RMSC 4002 course, completed by a group of six members.

[1]Code and replication data are available at *GitHub*.

Variable	Series
CPI	Consumer Price Index
Core CPI	Consumer Price Index less food and energy
PCE	Personal consumption expenditures
Retail Sales	Retail trade, excluding food services
Durable Goods Orders	New orders for durable goods
UNP	Unemployment rate
Wage Growth	Median hourly wage growth, nominal
PPI	Producer Price Index, all commodities
Industrial Production	Industrial production index
Trade Balance	Goods and services balance, USD million

Table 1: Macroeconomic indicators.

Correlation Figure 1 shows the correlation matrices for sector prices and macroeconomic data. Of the 55 pairs of sector prices, 85.5% have a correlation above 0.5. Energy is the least correlated sector, with correlations ranging from -0.08 with communication services to 0.53 with financials, possibly because it depends heavily on the crude oil market, which moves largely independently of the other sectors. The macroeconomic indicators are not uniformly positively correlated. Of their 45 pairs, 55.6% are positive, and 60.0% of the positive pairs exceed 0.5. CPI is strongly correlated with Core CPI (0.79) and Wage Growth (0.80), and PCE with Retail Sales (0.81) and Industrial Production (0.70). A plausible mechanism is that rising consumer demand increases personal consumption and retail sales, and firms respond by expanding output, which raises industrial production.

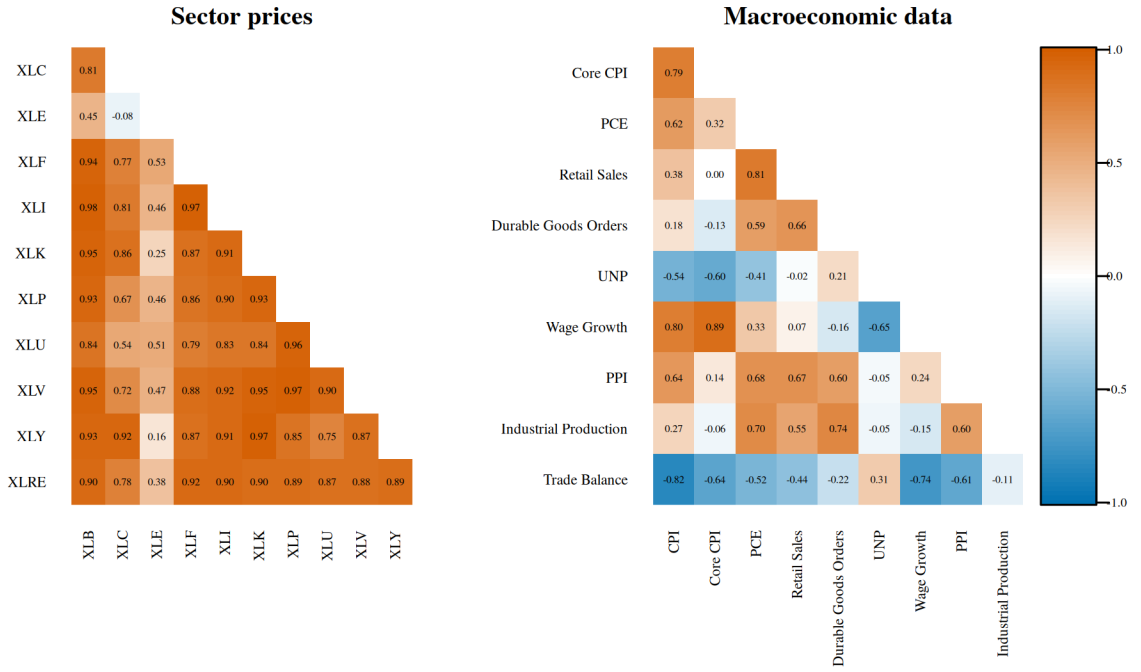


Figure 1: Correlations of the sector prices (left) and of the macroeconomic indicators (right).

3 Methodology

3.1 Principal component analysis

Let $X \in \mathbb{R}^{n \times p}$ and let Z be the standardized variables $z_{tj} = (x_{tj} - \bar{x}_j)/s_j$. The correlation matrix and its eigendecomposition are

$$R = \frac{1}{n-1} Z^\top Z = V \Lambda V^\top, \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p), \quad \lambda_1 \geq \cdots \geq \lambda_p, \tag{1}$$

where the columns $\boldsymbol{v}_k$ of V are the loadings. The kth component is $y_{k,t} = \sum_j v_{jk} z_{tj}$, with variance λ_k, and the components are mutually uncorrelated.

3.2 Selection of components

Three rules determine how many components to retain. The Kaiser rule retains component k when $\lambda_k > 1$, i.e., when it explains more variance than a single standardized variable. The scree plot (Cattell, 1966) plots λ_k against k, and the components before the eigenvalues level off are retained. The Pareto plot shows the cumulative share of variance, of which the retained components should capture at least 75%.

3.3 Principal component regression

Let r_t denote the S&P 500 log return and $y_{k,t}$ the kth component of the 11 standardized sector log returns. Model m regresses the index return on the first m components by ordinary least squares,

$$r_t = \beta_0 + \sum_{k=1}^{m} \beta_k \, y_{k,t} + \varepsilon_t, \qquad m \in \{1, 2\}. \tag{2}$$

In the testing sample, we compute the components using the training means, standard deviations, and loadings, and the error is $e_t = r_t - \hat{r}_t$. The number of components is also assessed by cross-validation over ten random segments of the training sample (Mevik and Wehrens, 2007), which reports the root mean squared error of prediction

$$\operatorname{RMSEP}(m) = \sqrt{\frac{1}{n} \sum_{t=1}^{n} \left(r_t - \hat{r}_t^{(-t)}(m) \right)^2}, \tag{3}$$

where $\hat{r}_t^{(-t)}(m)$ is fitted without the segment containing t.

4 Principal Components

Table 2 reports the leading eigenvalues. In both datasets, the first two eigenvalues exceed one and the third does not. Figures 2 and 3 show that the eigenvalues level off after the second component and that the first two components explain 78.26% of the macroeconomic variance and 93.62% of the sector price variance. The Kaiser rule and the scree plot retain two components in each dataset, and two components clear the 75% Pareto threshold in both.

4.1 Macroeconomic components

Figure 4 shows the loadings. The first PC accounts for most of the variation in the headline and core indicators. Its loadings on the inflation and consumption measures range from 0.22 to 0.41, led by CPI at 0.41, PCE at 0.39 and PPI at 0.35, while UNP and Trade Balance load negatively at -0.22 and -0.37. Although the loadings differ in size, they reveal a parallel movement of inflation and consumption. Under normal conditions, inflation erodes the purchasing power of consumers

	Eigenvalue			Cumulative share (%)		
	PC1	PC2	PC3	PC1	PC2	PC3
Macroeconomic data	4.85	2.97	0.86	48.52	78.26	86.86
Sector prices	9.04	1.26	0.38	82.14	93.62	97.07

Table 2: Leading eigenvalues and cumulative share of variance.

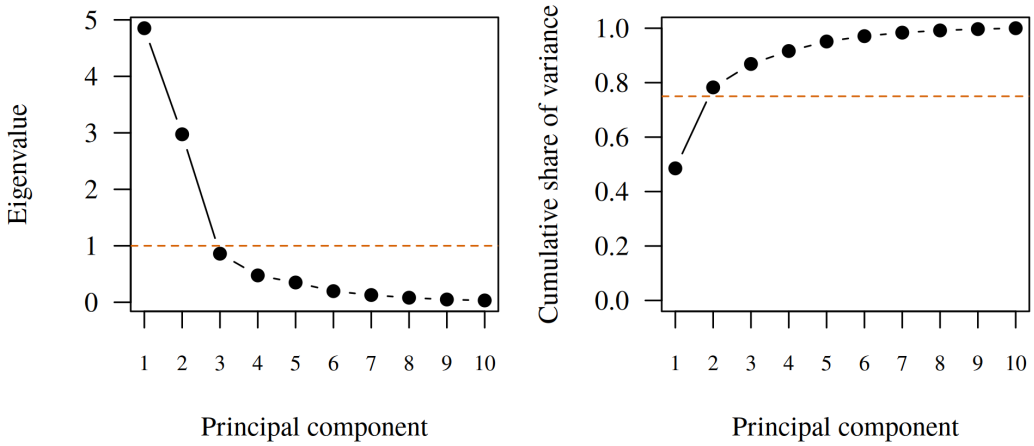


Figure 2: Scree plot (left) and cumulative share of variance (right) of the macroeconomic data.

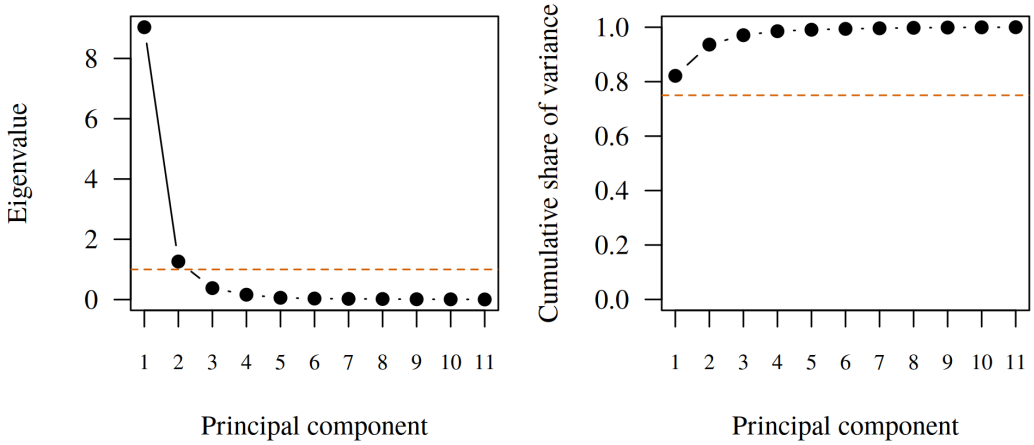


Figure 3: Scree plot (left) and cumulative share of variance (right) of the sector prices.

and reduces discretionary spending, so that consumption weakens when inflation is high. The first PC instead shows a positive association, which we attribute to the savings accumulated during the COVID-19 pandemic and to the popularization of a "You Only Live Once" (YOLO) attitude toward consumption. Lockdowns, reduced social activity and travel restrictions constrained the spending of many households during the pandemic. At the same time, the YOLO attitude, spread through social media and the influence of celebrities, favored immediate gratification over concern for future consequences. The excess savings of the pandemic, combined with this attitude, raised spending on discretionary items, experiences and luxury goods even as inflation rose.

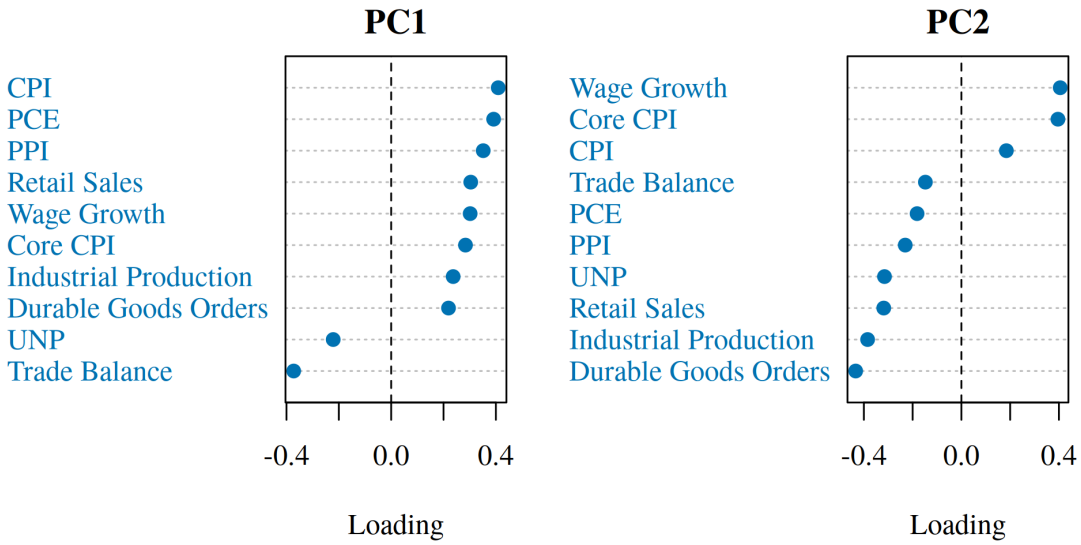


Figure 4: Loadings of the first (left) and second (right) macroeconomic components.

The second PC contrasts Wage Growth at 0.41 and Core CPI at 0.40 with Durable Goods Orders at -0.43 and Industrial Production at -0.38. The co-movement of durable goods orders and industrial production reflects energy demand, which operates through production costs. Durable goods orders measure demand for long-lasting goods such as appliances, automobiles, and machinery, and industrial production measures output in manufacturing, mining, and utilities. Both activities require substantial energy to power machinery and run manufacturing processes.

The positive loadings of Wage Growth and Core CPI follow from wage-price dynamics. Workers demand higher wages to preserve their purchasing power as prices rise, and higher wages in turn raise production costs, which pass through to consumer prices. The labor market also remained tight during the Federal Reserve's rate-hiking cycle, with unemployment near historical lows, which strengthened workers' bargaining power. The negative loadings of Durable Goods Orders and Industrial Production are consistent with consumers devoting a larger share of their budgets to essential expenses during inflationary periods, which reduces the demand for durable goods and, in turn, industrial production.

4.2 Sector components

Figure 5 shows the loadings. The first PC accounts for most of the variation in every sector except energy, with loadings between 0.27 and 0.33, while energy loads at 0.14. The loadings describe a parallel shift that aligns with overall market sentiment. When the economic outlook is optimistic, investors allocate capital across a wide range of sectors, and they withdraw from many sectors at once amid economic uncertainty and recession fears. News on growth, corporate earnings, policy, geopolitical tension or regulation shifts sentiment across sectors in the same direction. The

magnitude of the loadings also reflects each sector's resilience. Communication services and utilities, treated here as defensive sectors, load less, at 0.27 and 0.30, which is consistent with their stable demand, steady earnings and dividend payments. These features made them more resilient during recent economic slowdowns.

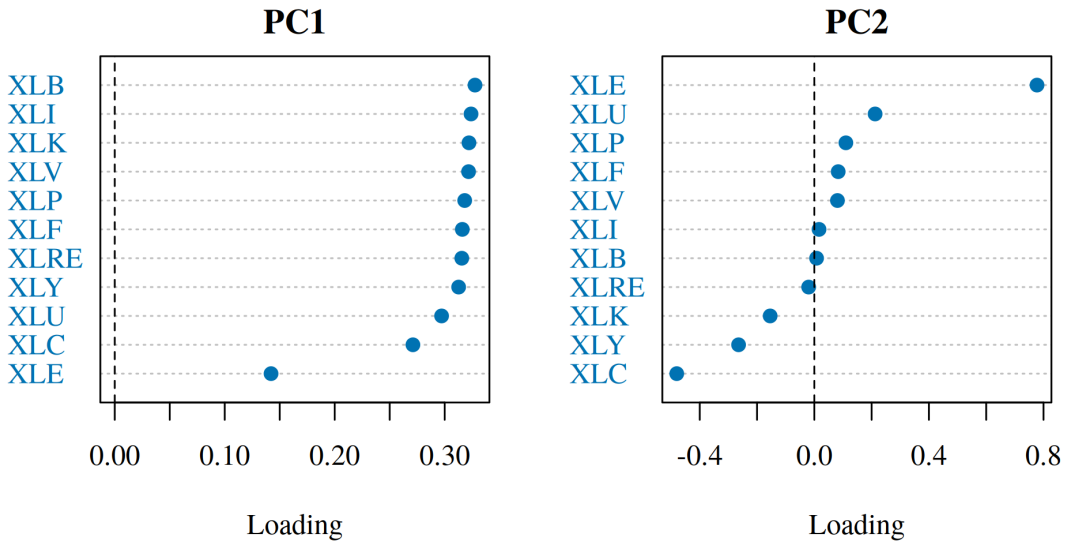


Figure 5: Loadings of the first (left) and second (right) sector components.

The second PC is an energy factor, with a loading of 0.78 on energy, while the loadings on the other sectors range from -0.48 for communication services to 0.21 for utilities. When energy moves, the other sectors remain largely unchanged or move slightly in the opposite direction. This pattern reflects supply instability and strong demand expectations in the oil market. Following the Russian invasion of Ukraine in February 2022, Brent crude rose to about USD 139 per barrel in March amid concerns about supply from one of the world's largest oil producers, and the Organization of the Petroleum Exporting Countries and its allies (OPEC+) kept to their planned monthly increase rather than raising production faster. The end of China's zero-COVID policy at the close of 2022 raised expectations of an economic rebound and stronger energy demand. Although the recovery fell short of expectations, OPEC+ announced deep production cuts from late 2022 that kept oil prices elevated through 2023. Selling at these prices while fixed costs remained stable, the energy sector expanded margins and raised dividends, attracting income-oriented investors.

4.3 Macroeconomic components and sector prices

The loadings suggest a link between the two sets of components, which this study examines qualitatively rather than by estimation. The market-wide sector component corresponds to the macroeconomic conditions that shape market sentiment. Although the first macroeconomic PC shows inflation, consumption and wage growth rising together, unfavorable macroeconomic releases often coincide and signal a slowdown, to which investors respond by withdrawing capital from equities, placing downward pressure on most sector prices at once.

The energy component corresponds to the second macroeconomic PC, in which durable goods orders and industrial production move against wage growth and inflation. Sanctions on Russian oil and natural gas exports raised energy prices and, with them, the production and operating costs of energy-intensive sectors such as manufacturing, transportation and utilities. Firms scaled back production, and consumers facing higher energy prices spent less on durable goods, which lowered durable goods orders and industrial production. In this environment, investors turned to the energy

sector for its profitability and dividends, so that the energy sector advanced while the other sectors remained unchanged or declined slightly.

5 Principal Component Regression

Table 3 reports, for each number of components, the cumulative share of the sector return variance preserved by the components and the share of the S&P 500 return variance explained by the regression. The first component preserves 72.50% of the sector return variance and explains 96.19% of the index return variance. The second adds 8.10% of the sector variance but only 0.95% of the index variance. Since the gain from the second component is small, we compare Model 1 with Model 2 to assess whether the second component improves the regression or leads to overfitting.

Components	1	2	3	4	5	6	7	8	9	10	11
Sector returns	72.50	80.60	87.69	90.36	92.65	94.54	95.96	97.26	98.46	99.31	100.00
S&P 500	96.19	97.14	99.21	99.26	99.31	99.32	99.32	99.36	99.49	99.78	99.78

Table 3: Cumulative share of the sector return variance preserved and of the S&P 500 return variance explained, in percent.

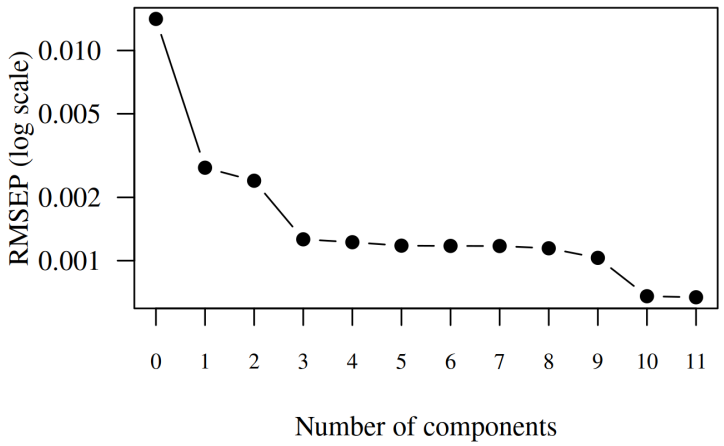


Figure 6: Cross-validated RMSEP against the number of components, on a logarithmic scale.

Figure 6 reports the cross-validation. The RMSEP falls from 1.41% with no components to 0.277% with the first, 0.240% with the second and 0.126% with the third, so the first component delivers most of the reduction. Figure 7 shows the residuals, with a standard deviation of 0.239% under Model 2 versus 0.276% under Model 1, so Model 2's fitted values lie closer to the observed returns.

In the testing sample, both models track the index return closely, as Figure 9 shows. Figure 8 compares the errors. The root mean squared error is 0.206% under Model 2 against 0.266% under Model 1, and the errors range from -0.68% to 0.88% under Model 2 and from -0.77% to 1.02% under Model 1. Model 2 also concentrates its errors near zero, with 72.0% of testing days within 0.2% of the actual return, compared with 56.4% under Model 1. Together with the training results, these findings indicate that Model 2 is the better of the two models, i.e., adding a second PC improves performance.

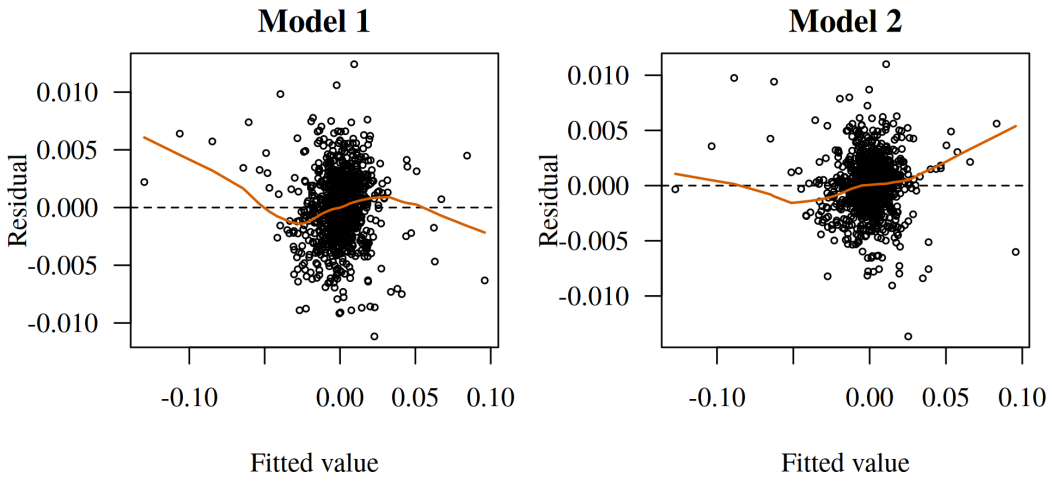


Figure 7: Residuals against fitted values. The solid line is a locally weighted smoother.

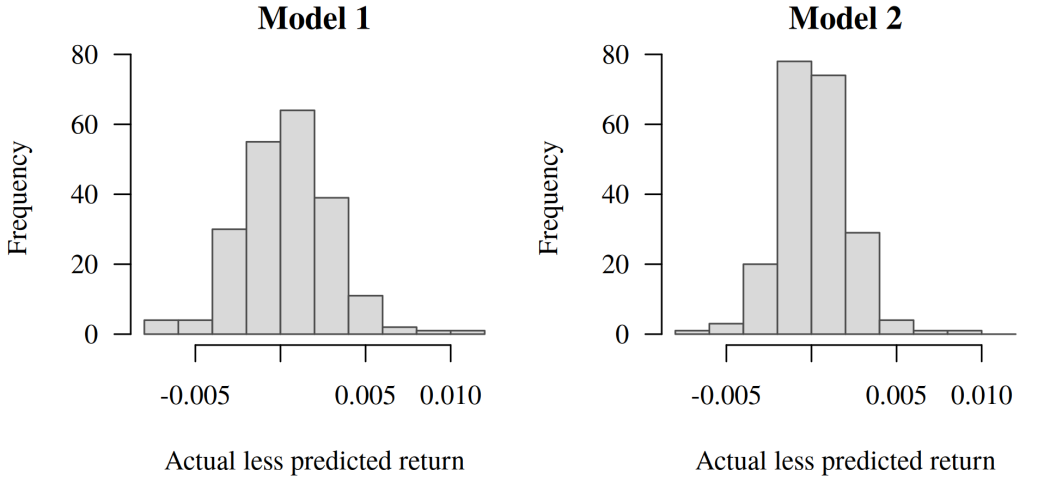


Figure 8: Distribution of the testing errors, defined as the actual less the predicted S&P 500 return.

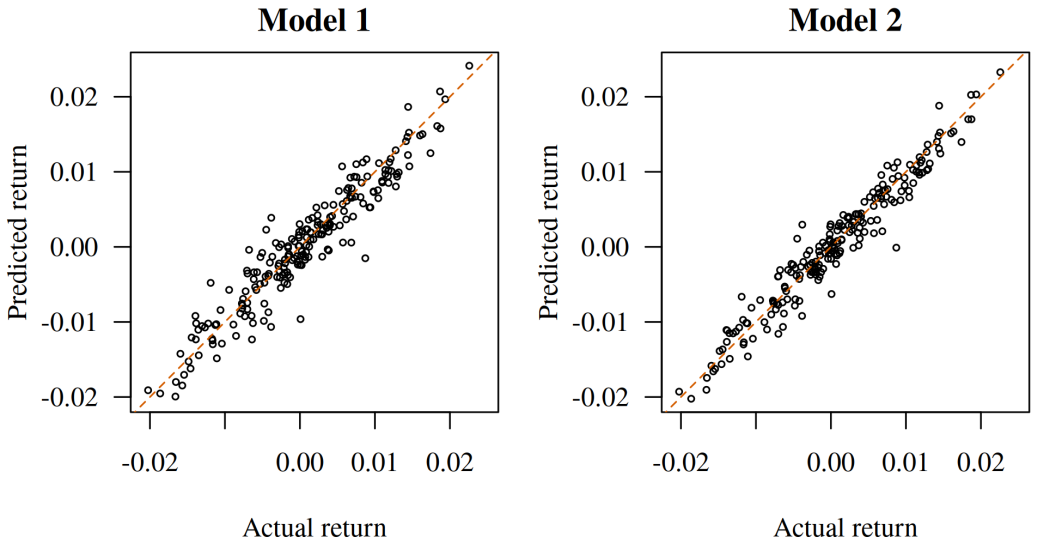


Figure 9: Predicted against actual S&P 500 log returns over the testing sample.

6 Conclusion

This study applies PCA to 10 US macroeconomic indicators and 11 sector prices, and two components summarize each dataset. The macroeconomic components describe a joint movement of inflation and consumption, which we attribute to pandemic savings and the YOLO attitude, and a contrast between wages and core inflation on the one hand and durable goods orders and industrial production on the other. The sector components describe a market-wide factor driven by sentiment and an energy factor driven by supply instability and robust demand expectations in the oil market. A PCR on the first two sector components explains 97.14% of the variance of the S&P 500 return and outperforms the regression on the first component in the testing sample, with a root mean squared error of 0.206% against 0.266%. It should be noted that the relationship between macroeconomic and sector components is analyzed only qualitatively, and future research could explore a quantitative approach.

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